13. Among the following cells

I) Leclanche cell

II) Nickel – Cadmium cell

III) Lead storage battery

IV) Mercury cell

Primary cells are

a) I and IV b) I and III c) III and IV d) II and III

14. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because

a) Zinc is lighter than iron

b) Zinc has lower melting point than iron

c) Zinc has lower negative electrode potential than iron

d) Zinc has higher negative electrode potential than iron

15. Assertion : pure iron when heated in dry air is converted with a layer of rust. Reason : Rust has the composition $Fe_3O_4$

a) if both assertion and reason are true and reason is the correct explanation of assertion.

b) if both assertion and reason are true but reason is not the correct explanation of assertion.

c) assertion is true but reason is false

d) both assertion and reason are false.

16. In $H_2$-$O_2$ fuel cell the reaction occurs at cathode is

a) $O_2(g) + 2H_2O\ (l) + 4e^- \rightarrow 4OH^-\ (aq)$

b) $H^+(aq) + OH^-\ (aq) \rightarrow H_2O\ (l)$

c) $2H_2(g) + O_2(g) \rightarrow 2H_2O\ (g)$

d) $H^+ + e^- \rightarrow \frac{1}{2} H_2$

17. The equivalent conductance of $\frac{M}{36}$ solution of a weak monobasic acid is 6 mho $cm^2$ equivalent $^{-1}$ and at infinite dilution is 400 mho $cm^2$ equivalent $^{-1}$. The dissociation constant of this acid is

a) $1.25\times10^{-6}$ b) $6.25\times10^{-6}$ c) $1.25\times10^{-4}$ d) $6.25\times10^{-5}$

18. A conductivity cell has been calibrated with a 0.01M, 1:1 electrolytic solution (specific conductance ($\kappa = 1.25\times10^{-3}$ S $cm^{-1}$) in the cell and the measured resistance was 800 $\Omega$ at 25°C. The cell constant is,

a) $10^{-1}$ c $m^{-1}$ b) $10^{1}$ c $m^{-1}$ c) 1 c $m^{-1}$ d) $5.7\times10^{-12}$



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